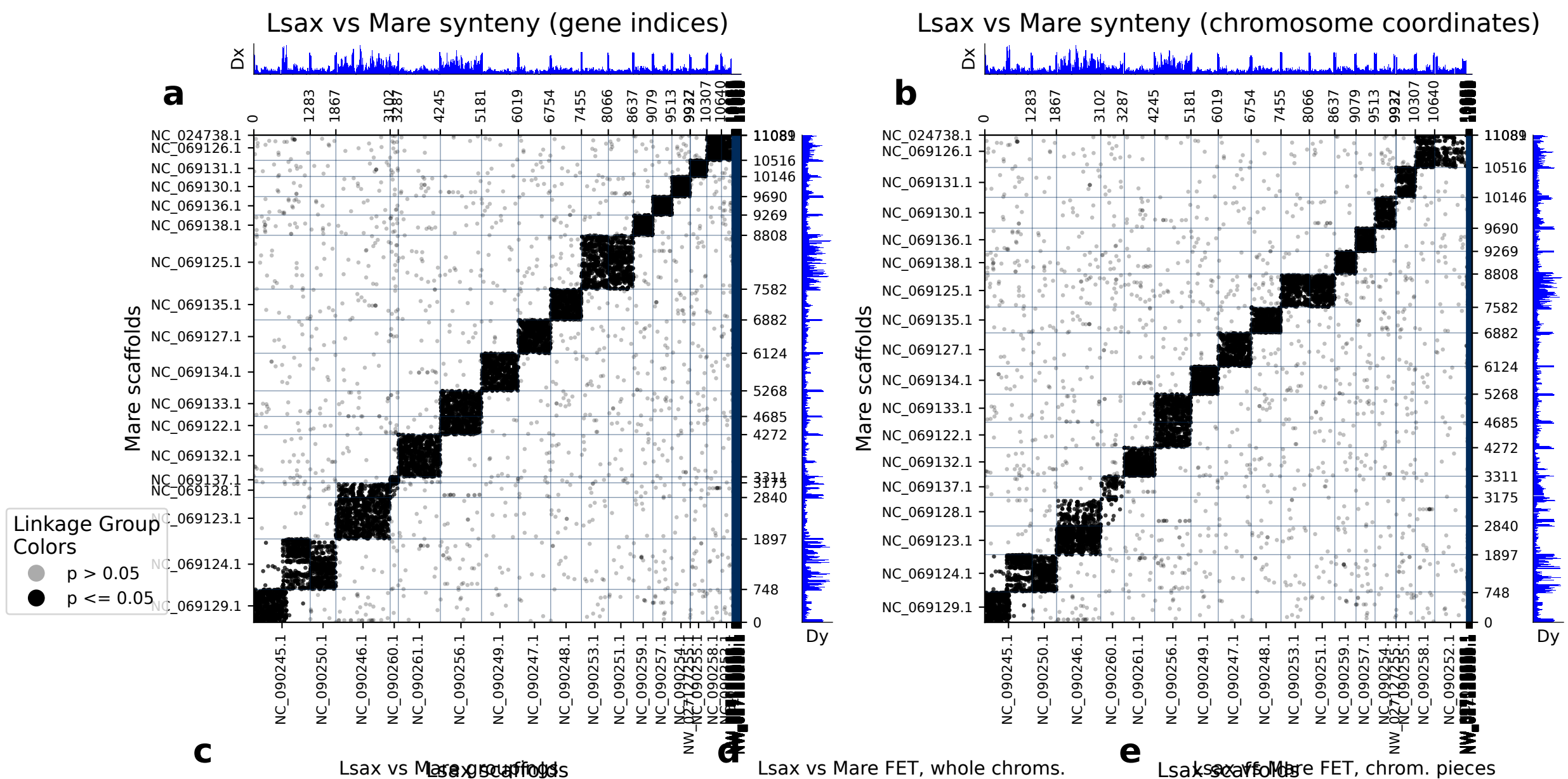


diamond: 11090 orthologs on 207 Lsax scaffolds and 18 Mare scaffolds.



Q Lsax vs Mare FET, whole chroms.

	Lsax scaf	Mare scaf	p (FET)	Count
0	NC_090246.1	NC_069123.1	0.00e+00	858
1	NC_090261.1	NC_069132.1	0.00e+00	850
2	NC_090249.1	NC_069134.1	0.00e+00	760
3	NC_090247.1	NC_069127.1	0.00e+00	657
4	NC_090245.1	NC_069129.1	0.00e+00	632
5	NC_090248.1	NC_069135.1	0.00e+00	606
6	NC_090253.1	NC_069125.1	0.00e+00	531
7	NC_090250.1	NC_069124.1	0.00e+00	513
8	NC_090251.1	NC_069125.1	0.00e+00	510
9	NC_090245.1	NC_069124.1	4.67e-186	494
10	NC_090256.1	NC_069133.1	0.00e+00	486
11	NC_090259.1	NC_069138.1	0.00e+00	368
12	NC_090257.1	NC_069136.1	0.00e+00	356
13	NC_090254.1	NC_069130.1	0.00e+00	354
14	NC_090256.1	NC_069122.1	0.00e+00	350
15	NC_090255.1	NC_069131.1	0.00e+00	295
16	NC_090258.1	NC_069126.1	0.00e+00	281
17	NC_090246.1	NC_069128.1	1.25e-148	236
18	NC_090252.1	NC_069126.1	1.86e-172	170
19	NC_090260.1	NC_069137.1	3.20e-118	84
20	NC_090260.1	NC_069128.1	1.08e-06	25
21	NC_030595.1	NC_024738.1	6.58e-25	8
22	NW_027127255.1	NC_069130.1	4.29e-04	5

	LSax scaffolds	Mare FET, chrom. pieces	
	LSax scaf	Mare scaf	p (FET) Count
0	NC_090246.1 : 0-end	NC_069123.1 : 0-end	0.00e+00 858
1	NC_090261.1 : 0-end	NC_069132.1 : 0-end	0.00e+00 850
2	NC_090249.1 : 0-end	NC_069134.1 : 0-end	0.00e+00 760
3	NC_090247.1 : 0-end	NC_069127.1 : 0-end	0.00e+00 657
4	NC_090245.1 : 0-end	NC_069129.1 : 0-end	0.00e+00 632
5	NC_090248.1 : 0-end	NC_069135.1 : 0-end	0.00e+00 606
6	NC_090253.1 : 0-end	NC_069125.1 : 0-end	0.00e+00 531
7	NC_090250.1 : 0-end	NC_069124.1 : 0-end	0.00e+00 513
8	NC_090251.1 : 0-end	NC_069125.1 : 0-end	0.00e+00 510
9	NC_090245.1 : 0-end	NC_069124.1 : 0-end	4.67e-186 494
10	NC_090256.1 : 0-end	NC_069133.1 : 0-end	0.00e+00 486
11	NC_090259.1 : 0-end	NC_069138.1 : 0-end	0.00e+00 368
12	NC_090257.1 : 0-end	NC_069136.1 : 0-end	0.00e+00 356
13	NC_090254.1 : 0-end	NC_069130.1 : 0-end	0.00e+00 354
14	NC_090256.1 : 0-end	NC_069122.1 : 0-end	0.00e+00 350
15	NC_090255.1 : 0-end	NC_069131.1 : 0-end	0.00e+00 295
16	NC_090258.1 : 0-end	NC_069126.1 : 0-end	0.00e+00 281
17	NC_090246.1 : 0-end	NC_069128.1 : 0-end	1.25e-148 236
18	NC_090252.1 : 0-end	NC_069126.1 : 0-end	1.86e-172 170
19	NC_090260.1 : 0-end	NC_069137.1 : 0-end	3.20e-118 84
20	NC_090260.1 : 0-end	NC_069128.1 : 0-end	1.08e-06 25
21	NC_030595.1 : 0-end	NC_024738.1 : 0-end	6.58e-25 8
22	NW_027127255.1 : 0-end	NC_069130.1 : 0-end	4.29e-04 5

(a) depicts the chromosomes/scaffolds of *Lsax* (x-axis) plotted against the chromosomes/scaffolds of *Mare* (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11090 orthologs found between 207 *Lsax* scaffolds and 18 *Mare* scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.